



6. Mental Resilience and Competitive Drive — Certain genetic factors influence dopamine and serotonin regulation, potentially affecting a player's ability to handle pressure, maintain focus, and thrive in high-stakes matches.

Do female tennis players have some anatomical, biomechanical, and physiological disadvantages compared to their male counterparts?

They do, and that can present certain disadvantages in performance. However, these differences do not diminish their athletic capabilities but rather shape how they train, compete, and excel in their own right.

Anatomical Differences

- **Hip Structure:** Women typically have a wider pelvis, which affects movement mechanics. This can slightly limit lateral speed and agility compared to men, whose narrower hips allow for more efficient side-to-side motion.

- **Shoulder and Upper Body**

Strength: Men generally have broader shoulders and a higher percentage of muscle mass in the upper body, providing more power in serves and groundstrokes.

- **Limb Proportions:** Women often have a higher Q-angle (the angle at which the femur meets the knee), which can influence knee stability and increase the risk of ACL injuries.

Biomechanical Differences

- **Serve Speed and Stroke Power:** Due to differences in upper-body muscle mass and leverage, men generally hit harder serves and groundstrokes. One of the fastest recorded women's serves (Sabine Lisicki at 131 mph) is still well below John Isner's record of 157 mph.

- **Explosiveness and Speed:** Men tend to have higher levels of type II (fast-twitch) muscle fibres, contributing to quicker bursts of speed and acceleration on the court.

- **Grip Strength and Racket Control:** Greater grip strength in men allows for better control over spin and shot variety, particularly on heavy topspin groundstrokes and

Short and swift: While taller players have powerful serves, shorter players, such as 5'7" Diego Schwartzman, typically have superior agility and baseline movement.

when returning powerful serves or volleying high-speed passing shots.

Physiological Differences

- **Aerobic and Anaerobic Capacity:** Men have a higher VO₂ max (oxygen uptake capacity), allowing them to sustain higher intensities for longer durations. However, women are often better at endurance due to differences in fat metabolism.

- **Hormonal Impact:** Oestrogen can lead to greater fat storage, affecting body composition, while testosterone in men promotes muscle hypertrophy and recovery.

- **Recovery and Injury Risk:** Women have higher rates of knee injuries (e.g. ACL tears) due to anatomical and hormonal factors. However, they often show greater resilience in endurance recovery.

How Elite Female Tennis Players